

Survivable Routing for WDM Networks: Challenges faced by the South African Service Providers

B. Tonderayi Mangara, F. Wilhelm Leuschner, *Pr. Eng., SMIEEE, MSAIEEE, FSAE*

Abstract—This paper discusses survivable routing for Wavelength Division Multiplexing (WDM) networks. The downside of boosting network capacity through a technology like WDM is increasing network vulnerability. With more than one network operator in South Africa network survivability will be one of the most important and challenging quality of service (QoS) criteria. This paper investigates new network survivability requirements, which need to be incorporated into the network design so that the network will remain connected after a fiber link or node fault.

Index Terms—bandwidth, optical networks, protection, restoration, WDM.

I. INTRODUCTION

THIS paper discusses survivable routing in WDM networks. The most important factors that have made optical fiber the broadband transmission medium of choice in telecommunications are higher capacity, higher reliability, longer repeater spacing, greater security, smaller size and less weight, unlimited growth potential, lower system costs, and scalability. The fiber optic medium is the only one capable of providing high-bandwidth service cost-effectively and today, it is widely deployed in backbone networks. However, only a small fraction of the full capacity of the installed optical fiber was realized until WDM was proposed to utilize the fiber capacity more efficiently. In WDM networks, a single fiber carries many channels, and the total utilized capacity can be increased dramatically.

WDM has reduced the mismatch between peak processing and source rates, in the order of several Gigabits per second and optical fiber capacity/transmission rates, which approach Terabits per second, by partitioning the optical fiber bandwidth into a few tens of disjoint wavelength bands. Each wavelength band is capable of operating at a more manageable rate up to 40 Gigabits per second.

WDM technology has enabled wavelength routing, which provides the benefits of reduced switching costs as compared to high-speed electronic switches and more transparency than electronics could provide. Also WDM needs only one optical amplifier whereas Spatial Division Multiplexing (SDM) (different fibers) requires one amplifier per fiber. Thus, WDM technology is paving a way for the development of new services and is playing a major role in

the expansion of optical networks.

It is also clear that the South African service providers need, with the impending introduction of the second and even third fixed network operator, to ascertain how to use WDM in order to utilize fiber capacity more efficiently as the bandwidth demand in South Africa is not as high as in, for example Europe.

While WDM creates an increase in the exploited bandwidth of available fibers by providing parallel channels in a single fiber, providing large bandwidth to users, and more revenue to service providers, it also has some potential problems. Since a large amount of traffic can be carried via a single fiber, a single fiber failure in the system may result in disconnection of a large number of user's end-to-end connections. Therefore, a survivable WDM network design is mandatory.

II. SURVIVABILITY MECHANISMS IN WDM OPTICAL NETWORKS

Network survivability, which is the ability of a network to withstand and recover traffic affected by failures, is one of the most important requirements in the design of telecommunication networks. Network survivability mechanisms guarantee seamless communication services in the face of network failures. Survivability techniques that have been proposed and used in WDM optical networks can be classified under two general categories: survivable network design/predesigned protection and dynamic restoration schemes. Survivable routing is an approach to allocate the spare capacity for a group of working lightpaths existing in a network with a goal of improving the capacity efficiency. A specific routing is known as survivable if the failure of any physical link leaves the (logical) network connected.

In designing schemes for survivable routing and spare capacity allocation there are many factors involved. Some of the most important ones are: resource utilization, request blocking ratio, recovery granularity, control complexity, tolerance of single/multiple faults and scalability. The final purpose or aim of the design is to achieve maximum resource utilization. However, it is difficult to attain all these goals at the same time and hence, trade-offs between different solutions are necessary (e.g. dedicated protection schemes offer faster recovery than restoration schemes, in most cases, but are less resource-efficient on the other hand). The trade-off between network survivability and network utilization is optimized if survivability techniques are adapted to the changing state of network resources, such as bandwidth.

In this paper we discuss survivability options and analyze

their respective properties with respect to some evaluation criteria that are of particular importance for South African service providers.

A. Multilayer Survivability in WDM Networks

The WDM layer supports different higher-layer services, for example Synchronous Optical Network (SONET) connections, Asynchronous Transfer Mode (ATM) virtual circuits, and IP-switched datagram traffic. With this layered structure network, survivability can be offered at the WDM layer or higher layers. A layered network offers the network designer a range of survivability options. A strategic decision has to be taken between incorporating survivability mechanisms at higher or lower network layers, or indeed at multiple layers. There is no simple algorithm to solve this problem. Johnson [9] gives some general principles on survivability options in layered networks. Providing resilience at multiple layers leads to the issues of assigning functions to each layer and coordinating the layers in effecting the recovery from a fault. Demeester [10] proposed two *escalation* or *interworking* (i.e. set of rules describing the point of origination of the fault recovery process and the interaction between the various layers) strategies based on the layer at which the fault recovery is initiated and also reported on a cost-comparison of the strategies.

B. WDM-Layer Survivability

With the advent of WDM systems being widely deployed in the backbone network, the use of optical switches and all-optical components introduces a new network layer, called the *optical* or WDM layer, into the layered network architecture. An important question for a multilayer network, as highlighted in the previous section, is in which layer to provide network recovery. A sensible possibility is to provide resilience in the optical layer. Providing survivability at the optical layer has the following advantages:

Speed – recovery at the optical layer is much faster, as the nodes can act quickly upon occurrence of failures and do not have to wait for higher-layer indication signals.

Transparency – the wavelength routing technique is independent of protocols used in higher layers.

Effectiveness – optical restoration makes more efficient use of restoration capacity because of resource sharing among different service layers.

C. Survivability Schemes in WDM Mesh Networks

In survivable routing the main objective is to route the lightpaths (corresponding to the links of a logical topology) in the physical topology in such a way, that failure of any single physical (optical fibre) link or node cannot disconnect the logical network.

Survivability is approached in two ways to recover from failure:

- Preplanned protection, and
- Dynamic restoration.

In *protection*, spare capacity is reserved during call set-up. In *restoration*, the spare capacity that is available after the fault's occurrence is utilized for re-routing the disrupted connections. In the following sections we summarize the

different schemes in WDM mesh networks.

1) Path-Based Protection/Restoration

Path-based protection entails the rerouting of all working lightpaths that use the failed link along precomputed backup lightpaths. Here, the entire route of the working lightpaths may be changed. This flexibility in path protection could lead to lower protection capacity but requires that all failed paths effect the recovery independently. In path protection, the protection capacity may be dedicated to a path or may be shared. In path restoration, when a fiber fails, the upstream end node of the failed fiber sends an alarm message to the source node of the disrupted optical label-switched path (LSP), while the downstream end node sends a tear down message to the destination node of the optical LSP. On receiving the alarm message, the source node tries to re-establish the optical LSP by sending a setup message to the destination node via an alternate path. Note: the end node of the failed fiber that is closer to the source (destination) node is called the upstream (downstream) end node.

a) Dedicated-Path Protection

Here, a protected lightpath is realized by two optical trails, called the working and protection paths. The working and protection paths do not have a common point of failure. The working path carries the lightpath signal under normal operating conditions, and the protection path carries the signal when there is a failure along the working path. Dedicated protection is commonly referred to as 1+1 and 1:1 protection. In 1+1 protection, two disjoint paths are used and data is sent on both paths. The receiver picks the better path to use and discards data from the other. In 1:1 protection, two disjoint paths also exist, but data is sent only on one active path.

b) Shared-Path Protection

At the time of call setup for a primary working path, a link-disjoint backup path and wavelength are also reserved. However, the backup wavelength reserved on the links of the backup path may be shared with other backup paths. Thus, it is possible to utilize the capacity more efficiently, while still achieving hundred percent recovery from single-link failures.

c) Path Restoration

Here demands are shortest path routed and just enough spare capacity placed to support multi-commodity maximum-flow type end-to-end re-routing of all simultaneously failed working paths. A centralised or distributed re-routing mechanism deploys a collectively coordinated set of replacement paths in response to the specific failure. Surviving portions of the failed working paths can be re-used for restoration.

2) Sub-path Restoration

When a fiber fails, the upstream end-node tries to patch the optical LSP by sending a setup message to the destination node. Meanwhile, the downstream end-node sends a teardown message to the destination node of the optical LSP.

3) Link-Based Protection/Restoration

In link protection, an alternate lightpath (called as a backup lightpath) between the end points of each link is *pre-computed*. Upon the link's failure, all the lightpaths using the link (called as working lightpaths) are switched at the end-nodes of the link to their corresponding backup lightpaths. The portion of the working lightpaths excluding the failed link remains the same. Link protection may require more protection capacity because of reduced flexibility in rerouting, requires only local knowledge around failed link to complete the recovery. In link protection, the protection capacity may be dedicated to a link or may be shared. In link restoration, when a fiber fails, the upstream end node does not send an alarm message to the source node; the downstream end node does not send a teardown message to the destination node; instead, the upstream end node tries to reroute the optical LSP around the failed link by sending a setup message to the downstream end node.

a) Dedicated Link Protection

In dedicated link protection, a protection wavelength path is dedicated to a working channel on a particular link. Hence, if the protection paths for (some wavelengths on) two different links overlap, different wavelengths must be assigned to the protection path on the overlapping portion even if the working wavelengths on the two links are the same. In general, it may not be possible to allocate a dedicated backup path around each link of the primary call, and on the same wavelength as the primary path. Therefore, designing efficient dedicated link protection schemes in large networks is difficult, especially if wavelength conversion facilities are unavailable. However, dedicated link protection may offer protection against the failure of multiple links.

b) Shared Link Protection

The basic idea of link-based protection was originally devised for ring-based network architectures such as SONET, where network-planning efforts significantly influence performance. Over the years link-based protection has migrated from ring-based networks to mesh networks. Generally, link-based protection in mesh networks is defined as follows: a protection mechanism that performs fault localisation during the occurrence of a failure, restores the interrupted services by circumventing the traffic from a failed link or node at the upstream neighbour node, and merges the traffic back to the original working path at the downstream neighbour node. From the above definition we see that, to protect both the downstream neighbour link and node, two merge nodes must be arranged for every node along a working path.

Shared link protection provides the fastest restoration due to fault localisation and better throughput due to relaxation of the shared risk link group (SRLG) constraint. The disadvantage of this mechanism is that the downstream neighbour node and link are required to have separate protection segments, which may impair performance by consuming extra protection resources. Thus, to perform a link-based shared protection on both links and nodes along a working path, a new scheme is required such that the consumption of protection resources is reduced without losing much restoration speed.

4) Dynamic Restoration

In general, *dynamic restoration* and *preplanned restoration* are the two major types of restoration schemes used in fixed networks. Dynamic restoration is more adaptive to network topology changes and traffic variance than preplanned restoration. The latter restoration mechanism offers a speed advantage against the former. Based on advantages and disadvantages of both types of schemes, broadband network designers should give much attention to hybrid restoration schemes.

D. Comparison of Survivability Mechanisms

Here we present a summary of the survivability mechanisms discussed.

TABLE 1: COMPARISON OF SURVIVABILITY MECHANISMS

WDM Survivability Mechanism		Resource Utilisation	Protection Switching / Restoration Speed	Recovery Ratio (Efficiency)	Ease of Control
Path Level	Dedicated Path Protection	Average	Excellent	Excellent	Excellent
	Shared Path Protection	Good	Good	Excellent	Average
	Path Restoration	Excellent	Poor	Good	Poor
Link Level	Dedicated Link Protection	*	*	Poor	*
	Shared Link Protection	Poor	Excellent	Excellent	Good
	Link Restoration	Excellent	Average	Average	Poor

*There was no comparative data available from the literature at the time of writing this paper.

III. CLOSING COMMENTS

Survivable routing is important because it provides continuous service in the presence of failures, especially link failure that is prevalent in South Africa (SA) because of cable thefts. Critical aspects of network design and planning to facilitate and increase survivability are:

- Reduced consumption of backup capacity whilst still maintaining a high degree of survivability.
- Fast and robust restoration.
- Efficient restoration.

Other issues in this regard, which need to be thoroughly investigated in the South African context, are:

- Capacity growth on the core network.
- Cost of optical network infrastructure.
- Expertise in survivable routing in SA.

To build resource efficient networks and higher margin services, carriers need not only look at the raw capacity but also at the intelligent delivery of that bandwidth. SA service providers need to fully understand that technology transitioning; rapid service delivery and end-to-end network intelligence are major carrier requirements.

ACKNOWLEDGMENT

We gratefully acknowledge the comments and contributions to this paper from Ms Ronelle Geldenhuys of the University of Pretoria.

REFERENCES

- [1] Zhang, H.; and Durrezi, A.; "Differentiated Multi-layer Survivability in IP/WDM Networks", IEEE/IFIP Network Operations and Management Symposium (NOMS) 2002, pp. 681-694.
- [2] Zhou, D.; and Subramaniam, S.; "Survivability in Optical Networks", IEEE Network, vol. 14, no. 6, Nov/Dec. 2000, pp. 16-23.
- [3] Qiao, C.; Datta, D.; Ellinas, G.; Gladisch, A.; and Modiano, E.; "Guest Editorial: WDM-Based Network Architectures", IEEE JSAC, vol. 20, no. 1, Jan. 2002, pp. 1-5.
- [4] Ye, Y.; Assi, C.; Dixit, S.; and Ali, M.A.; "A Simple Dynamic Integrated Provisioning/Protection Scheme in IP over WDM Networks", IEEE Communications Magazine, Nov. 2001, vol. 39, no. 11, pp. 174-182.
- [5] Modiano, E.; and Narula-Tam, A.; "Survivable Routing of Logical Topologies in WDM Networks", Proceedings of the INFOCOM 2001, Twentieth Annual Joint Conference of the IEEE Computer and Communications Societies, vol. 1, pp. 348-357.
- [6] Mukherjee, B.; "WDM Optical Communication Networks: Progress and Challenges", IEEE JSAC, vol. 18, no. 10, October 2000, pp. 1810-1824.
- [7] Ho, P. -H.; and Mouftah, H.T.; "Survivable Routing with SLSP Framework in Dynamic Optical Networks", IEEE Global Telecommunications Conference (GLOBECOM) 2002, vol. 2, pp. 1591-1597.
- [8] Modiano, E.; and Narula-Tam, A.; "Survivable Lightpath Routing: A new Approach to the Design of WDM-Based Networks", IEEE JSAC, vol. 20, no. 4, May 2002, pp. 800-809.
- [9] Johnson, D.; "Survivability Strategies for Broadband Networks", Global Telecommunications Conference (GLOBECOM) '96, Communications: The Key to Global Prosperity, 18-22 Nov. 1996, vol. 1, pp. 452-456.
- [10] Demeester, P. et al; "Resilience in Multilayer Networks", IEEE Commun. Mag., vol. 37, no. 8, Aug. 1999, pp. 70-76.

Tonderayi Mangara (M'00) received his T_1 and T_2 Diploma (Telecomm) from CGLI, London, UK, his Diploma in Electronics from Wuhan Radar Academy, Wuhan, Wubei, P.R. of China, his B.Sc. Eng. (Electrical) from the University of the Witwatersrand, Johannesburg, South Africa and his M.Eng (Electronic) from the University of Pretoria, Pretoria, South Africa. He is currently pursuing a PhD in optical networks at the University of Pretoria where he is a research and teaching assistant in the Dept. of Electrical, Electronic & Computer Engineering. His research interests include optical networks, transport network survivability, network topology, and network optimization.

Wilhelm Leuschner (Leuschner@eng.up.ac.za) is a consultant in optical networks, Chairman of the school of engineering, professor and head of department in Electrical, Electronic & Computer Engineering at the University of Pretoria, South Africa. He received his B.Sc. (Eng), his B.Eng (Hons), his M.Eng and his D.Eng from the University of Pretoria. His research interests include Photonics, lasers, optical fibers and fibre communication systems.